

Exam Date & Time: 14-May-2025 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

Data Analytics with Python [NPT-3201]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) **For the below given contingency table, if an employee of the company selected randomly, identify the probability of the employee as a female or a professional worker.**
- A)

Type of Position	Sex		Total
	Male	Female	
Managerial	8	3	11
Professional	31	13	44
Technical	52	17	69
Clerical	9	22	31
Total	100	55	155

(5)

- B) Given the dataset: [5, 7, 8, 10, 100], identify the best measure of central tendency and justify your choice. (5)
- 2) Explain the purpose of ANOVA in data analysis and the conditions under which it is appropriate to use it. Support your answer with a real-world example relevant to analytics. (5)
- A)
- B) A school principal believes that the average math score of students in her school is greater than the national average of 70. She asks the math department to test this claim using a sample of 40 students. Based on the above consequences, Identify the following:
- Null hypothesis (H_0) and alternative hypothesis (H_1) clearly.
 - The type of hypothesis test that is appropriate here: one-tailed or two-tailed. Justify your choice. (5)
 - Rejection of null hypothesis in this context.
- The decision the department should make, if the p-value obtained is 0.03 and the significance level is 0.05. What
- 3) Analyse how residual analysis is used to evaluate the performance of a regression model (5)

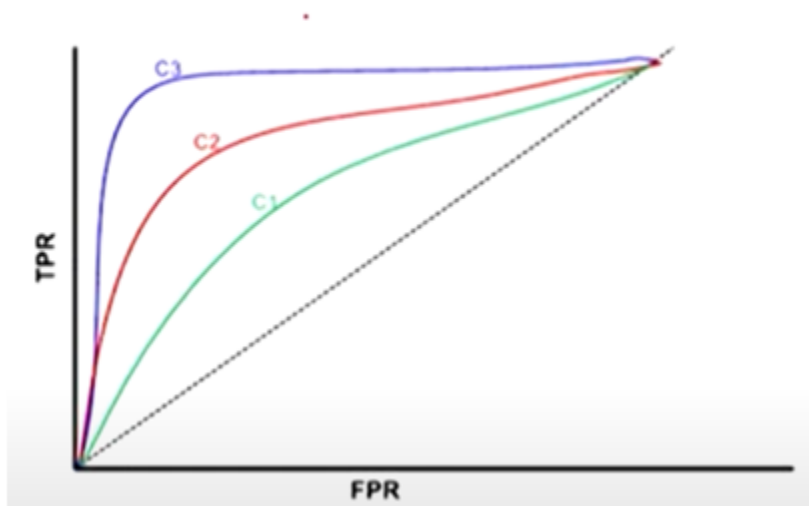
and improve prediction accuracy. Include the key patterns to look for in residual plots.

A)

B) Apply logistic regression to classify whether a student will pass or fail based on study hours and attendance. Also interpret the model's output. (5)

4) **For the given ROC curves for C1, C2 and C3, discover the best fit model using AUC.**

A)



(5)

B) Apply the Chi-Square goodness of fit test for uniform distribution with suitable example and explain your analysis (5)

5) Compare K-Means and Hierarchical Clustering in terms of their working principles, scalability, and use cases. (5)

A)

B) Define Probabilistic and non-probabilistic Sampling with their respective types and an example for each (5)

-----End-----